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Cigarette butt littering as a potential source of water pollution and human health risk

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ABSTRACT

Smoked cigarette butts (CBs) serve as reservoirs for harmful contaminants. The objective of the current study is to assess the CB littering in Aizawl city and estimate its potential contribution to water pollution in downstream surface water sources, along with the associated human health risks. The spatial-temporal distribution of CB littering density (CBD) across the city, spanning various land-use patterns, was analyzed through a visual-oriented littering survey. With an annual average CBD of 0.438 ± 0.669 CBs/m^2, commercial areas were identified as the littering hotspots. Further, the water pollution potential index (WPPI) was formulated considering leached concentrations of different contaminants from CBs into the water and their standard permissible limits. WPPI was calculated as 5.22×10^{-02}, which represented water pollution potential from a single CB/L of water. This corresponded to a cumulative WPPI of $(59.97 \pm 6.68) \times 10^{-06}$ from CB littering to surface water pollution through municipal sewerage system. The health risk assessment associated with CB littering was analyzed through the potential chronic daily intake (CDI) of leached contaminants, non-cancer risk (NCR), and cancer risk (CR). Heavy metals (lifetime exposure of 1125.9 mg from 1 CB/L) were found to be the most significant contributor to the cumulative CR (3.847×10^{-06}) and hazard index (HI) (1.785×10^{-02}). Further, it was estimated that the potential CR associated with 242 CBs/L in water is equivalent to smoking 1 cigarette per day for a lifetime. The WPPI, CR, and HI suggested that littered CBs can significantly contribute to water pollution.

1. Introduction

Approximately 7.41 trillion cigarettes were consumed by 1.14 billion smokers in 2019, and a significant portion (≈ 80 %) of these cigarettes, post-consumption are discarded as cigarette butts (CBs) [1–3]. CB littering is a significant environmental issue, frequently occurring in urban areas, public spaces, and beaches, making them one of the most littered items worldwide [4]. An estimated 4.5 trillion CBs are disposed of annually into the environment [5], contributing to around 1.2 million tonnes of CB waste generation [6]. CBs consist of cellulose acetate filter, wrapper paper, and residual tobacco [7]. The environmental impact of CBs is mainly attributed to the slow degradation of the filter (≈ 7.5–14 years) as a source of microfibre pollution [8–10] and the leaching of contaminants from the filter into water bodies [11]. Their small size and multi-point littering patterns result in widespread dispersion throughout the environment. Zafeiridou et al. [12] depicted that a single smoked CB can contribute to a water footprint of 3.7 L, a climate change impact of 14 g CO_2 equivalent emission, and 3.5 g oil equivalent to fossil fuel depletion.

The smoked CB filters serve as secondary sources of contaminants, releasing over 7000 hazardous chemicals including nicotine, heavy metal(oid)s (HMs), benzene, toluene, ethylbenzene, xylene (BTEX compounds), polycyclic aromatic hydrocarbons (PAHs), and aromatic amines (AAs) [6,13–17]. Green et al. [18] reported that nicotine present in 1 CB can contaminate up to 1000 L of water. The leached toxic substances can potentially contaminate surface/sub-surface water bodies through surface runoff and/or groundwater infiltration [19]. Moreover, the potential of CB filter-laden dissolved organic matter as a source of secondary pollutants was depicted by Parveen et al. [20]. The toxic effects of CB-laden water pollution on several freshwater and marine aquatic organisms, including fish [21–24], tadpoles [25], and snails [26], have been depicted by previous studies. Therefore, these contaminants have the potential to bioaccumulate and enter human food chains through biomagnification [27,28].

CB littering surveys effectively evaluate the litter abundance and spatial-temporal distribution, facilitating the identification of concentrated hotspots [29]. Previous studies have used various indices, including CB littering density (CBD) [30–32], cigarette butt pollution

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index (CBPI) [33–36], environmental status (ES) [37], clean coast index (CCI) [38], clean environment index (CEI) [39,40], and the Geographic Information System (GIS) approach [29] to estimate CB pollution within specific study areas. Most of these studies are concentrated in beaches [19,30,40,41], urban areas of plain terrains [29,31,32,34,42], and marine environments [35,41,43]. Among them, only CBPI assessed the potential leakage of pollutants from CBs to the soil and water, considering factors including CBD and the distance from groundwater [31]. However, the potential of CBs as a source of surface water contamination through runoff has not been studied before. This limitation is especially crucial for estimating water pollution in urban areas primarily reliant on surface water sources, like many hilly terrain cities. Especially in cities with inadequate wastewater collection infrastructure, the potential for CBs and their leached contaminants to enter downstream water bodies that serve as critical drinking water supply sources is much higher. Moreover, considering the hazardous nature of these contaminants, analyzing the potential human health risks from oral exposure to leached CB pollutants is important, which has never been reported before in previous literature. Therefore, the novelty of the current study lies in understanding the potential of CBs as a source of surface water contamination and assessing the human health risks associated with it.

The primary objective of this study was to assess CB littering in Aizawl, a hilly terrain city in Mizoram, India, and to evaluate its potential contribution to water pollution and the associated human health risks. The specific objectives of the current study were i) to identify annual average CB density and littering hotspots in the area through a multi-seasonal visual survey, ii) to analyze the impact of littered CBs on surface water quality by determining the water pollution potential index (WPPI) using models based on secondary data, and iii) to conduct human health risk assessment for potential chronic daily intake (CDI), non-cancer risk (NCR), and cancer risk (CR) associated with CB littering.

2. Methodology

2.1. Study area

The CB littering survey was conducted in Aizawl city (23.7307° N, 92.7173° E), the administrative capital of Mizoram state, in the north-eastern region of India (Fig. 1). The city is located in a hilly and mountainous terrain with an average elevation of 1132 m above sea level. Aizawl features humid sub-tropical climatic conditions with an annual average rainfall of 2246.3 mm [44] and has a net groundwater availability of 0.20 billion cubic meters (BCM) [45]. July was the wettest month (mean rainfall of 398.4 mm), while January and February had no precipitation. The mean temperature of the city varies from 16.7 to 27.8 °C, with January and May being the coldest and hottest months, respectively [44]. The city hosts a population of 400,309 with an average population density of 112 per km^2 [46]. The city demographics mainly comprise an indigenous tribal population with very high cigarette consumption rates. According to the Global Adult Tobacco Survey (GATS) 2017, the state of Mizoram, often called the “Cancer Capital of India” [47], has the highest prevalence of tobacco smoking in India (35 %) relative to the national average of 10.7 % [48]. Consequently, Aizawl district was also reported to have the highest age-adjusted cancer incidence rate in India, which was one out of every five people in the age group of 0–74 years [49,50]. Apart from the health effects of smoking, CB littering is also widely prevalent across the city (Figure S1). This was possibly contributed by a lack of CB collectables, no public smoking restrictions, and inadequate environmental awareness of the problem.

Currently, Aizawl depends exclusively on the Tlawng River, located about 12 km away from the city as its sole source of potable water. This is facilitated by the operation of a singular water treatment facility, which delivers 34 million liters per day (MLD) of treated water to the city [51]. Surface sewers collect the wastewater from the city along the curbside of the roads that drain into the Tlawng River, specifically upstream of the drinking water intake point of the city [52]. The city also

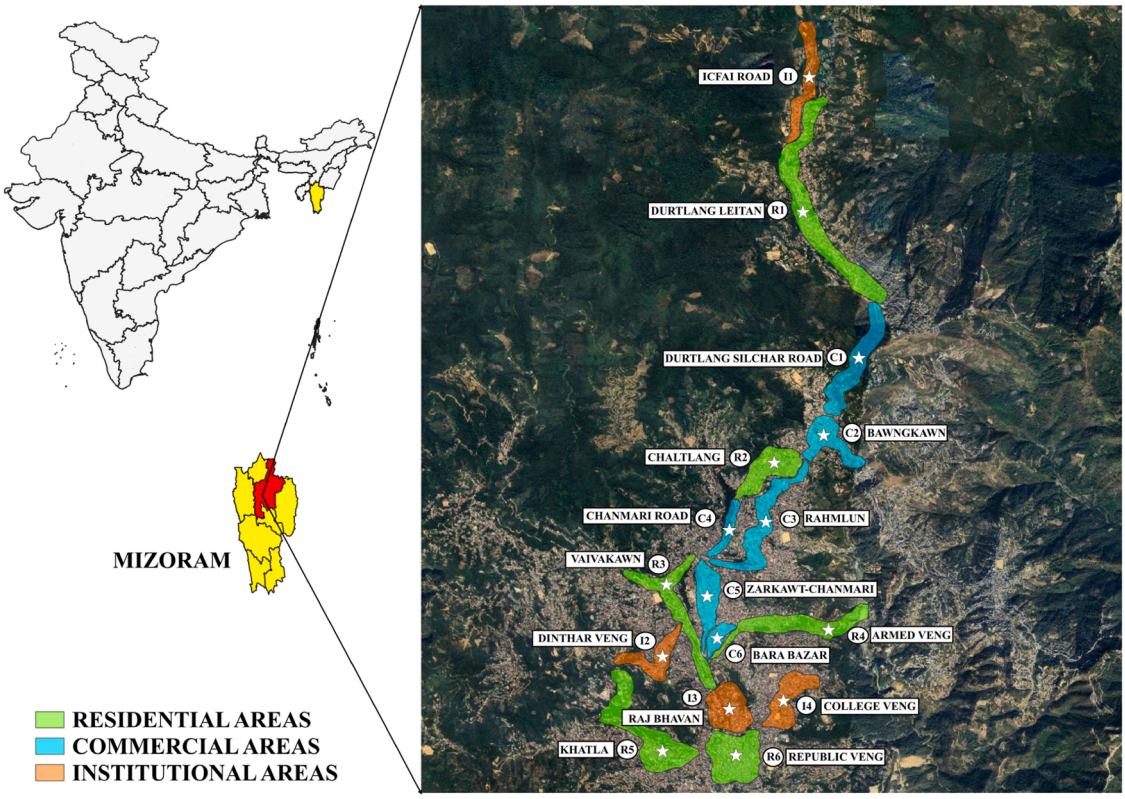


Fig. 1. Map of Aizawl City with study locations and their land use types.

hosts a recently constructed sewage treatment plant (STP) of 10 MLD capacity; however, owing to deficiencies in the sewerage collection and conveyance network, the operational efficiency of the STP is constrained.

2.2. CBs littering density survey

A visual-oriented litter survey methodology was adopted for this study [37,53], which was conducted from March 2023 to February 2024 (Fig. 1). Land use pattern is a significant factor that affects the CB littering patterns and densities [31,33,42,54]. So, a total of 16 locations were selected across the study area (Aizawl city), which were further categorized into institutional, residential, and commercial locations based on the land-use pattern [55] (Figure S2). As depicted in Table S1, the studied locations included four institutional (I1 – I4), six residential (R1 – R6), and six commercial (C1 – C6) locations. The study locations were strategically selected to ensure comprehensive coverage of all major roads and sections of Aizawl city and to capture the litter density variations influenced by land use. Additionally, the availability of surface drainage systems for runoff collection was a key criterion for maintaining consistency in the survey methodology across all littering points. These locations span the city's major urban centers, so they are all equipped with surface drainage systems.

The survey involved visual observation and counting of CBs littered along the road curbside adjacent to the sewers (encompassing both within the sewer and on sidewalks) on either side of the lane within a 1-meter span across the street [37]. All the locations were thoroughly surveyed to identify the points of CB littering, and the number of littered CBs found at each specific point was noted down. Considering the effect of environmental factors, including humidity and temperature on their weight, weight measurements were omitted from the scope of the study [33,40]. Since the samples were not collected, these points were designated as 'littering points.' The information regarding the average number of littering points (mean ± standard deviation (SD)) identified in each location was given in Table S1. The littering points in commercial and institutional areas were further categorized based on their specific land-use types to identify the hotspot land-use types where CB littering is prominent in the city.

Considering the influence of seasonal variations on the smoking behavior of individuals and spatial-temporal variation in the smoking and littering patterns of the population [32,33], the survey was conducted during the span of one year at all the locations with triplicate observations in three different seasons: winter (December to February), summer (March to May), and rainy season (June to September). The survey was conducted during the evening hours (3–5 PM) on the weekdays (Monday-Friday). This specific time was selected, considering it to be the peak time for the abundance of CBs on the street and to minimize the impact of street sweeping activities on littering density by considering the local street sweep timings (5–6 AM). Previous studies [29,39] have also reported that evening hours are the most appropriate time in their CB littering study.

The littered CBs were counted at each littering point, and their coordinates were recorded using the Marine Debris Tracker App. The mean of littered CBs observed at each location from the survey conducted across three seasons in triplicates was calculated. Further, the CBD at the studied locations was calculated by dividing the average number of littered CBs by the studied area (m^2) using Eq. 1 [56].

$$CBD = \frac{Number\ of\ CBs\ littered}{L \times W} \tag{1}$$

where, CBD is the cigarette butt density (unit/m^2), number of CBs littered is the count of CBs in the study area, L is the length (m) and W is the width (m) of the street.

The average CBD of all three seasons in individual studied locations and their spatial data were plotted as a heat map to identify the hotspot locations with the highest littering density.

2.3. Water pollution potential assessment

Since Aizawl city relies on surface sources to meet its water demands and the untreated discharge of sewers into nearby rivers, we considered the surface water pollution assessment rather than the frequently studied groundwater contamination. While analyzing groundwater contamination, studies have considered the E-coefficient which is a function of annual rainfall, soil characteristics, type of pathway surfaces, and groundwater level [42]. Since the present study aims to analyze the potential of surface water pollution from CBs, WPPI was formulated considering the leached concentrations of different contaminants from CBs into the water and their standard permissible limits (Eqs. 2 and 3).

$$PL_i = 1 + \left(\frac{C_i - S_i}{S_i}\right) \tag{2}$$

$$WPPI = \frac{1}{n}\sum_{i=1}^{n} PL_i \tag{3}$$

where, PL_i is the pollution load of the ith contaminant, C_i is the observed concentrations of the ith contaminant (μg/L), S_i is the standard permissible limit of the ith contaminant (μg/L), *WPPI* is the summation of pollution loads for all the contaminants (unitless), and n is the number of contaminants considered.

Hossain and Patra [57] used a similar methodology to calculate the water pollution index (WPI) to assess surface/groundwater quality. The WPI analyzed the water pollution load of groundwater collected from sample locations as a measure of the water quality status of the region. In contrast, the WPPI adopted in the current study considers the water pollution potential of a single CB/L of water. This was further used to calculate the cumulative pollution potential of all CBs littered across Aizawl city to the surface water through the city sewerage system. The WPPI calculated for a single CB can be widely applied to different locations to give a general idea of the water pollution potential from CB littering. This index can hold particular significance in areas where CB littering occurs near water bodies.

The concentration data of CB leachate were obtained from comprehensive literature reviews of past studies (Table 2). Various bibliographic databases including Scopus, Web of Science, and Google Scholar were utilized with keywords 'leachates', 'contaminants', 'toxic compounds', 'pollutants', 'heavy metals', 'PAHs', 'BTEX', and 'Aromatic Amines'. A total of 62 CB-related studies were identified that meet our criteria, out of which 26 studies reported data on HMs leaching and 14 studies specifically examined the leaching of organic contaminants. Based on the available literature on the analysis of contaminants leached from CBs, the current research was limited to using deionized water as the leachate medium for CBs. However, it should be noted that the leachability characteristics of CBs can be influenced by water characteristics like pH, ionic strength, and salinity, as well as environmental factors such as temperature and sunlight, which can be the focus of future research. Preference was given to studies published in high-impact Q1 journals, while other studies were excluded from consideration. The WPPI was calculated using these concentrations as input values (C_i in Eq. 2). Since concentrations were often presented as ranges (mean ± SD), Monte Carlo simulation was employed. Monte Carlo simulation is a powerful tool for modelling complex systems and uncertainty, allowing for generating probabilistic outcomes based on random sampling [58]. This involved considering each concentration range's minimum, maximum, and most likely (mode) values. To determine these values, concentration ranges from various studies were utilized to generate 10,000 random datasets using the Random Data Generator tool in Minitab software. When multiple studies reported the same contaminant and concentration, each range was separately subjected to random generation before being amalgamated to ascertain the minimum, maximum, and mode values.

Given that the leaching rate and metal concentration from CBs are contingent upon factors such as the degradation stage of the CBs, as well

as environmental parameters like humidity and duration [59], the range of concentrations incorporated into the model was presumed to account for this variability. Further, in our assumptions, we considered that the CBs scattered in and around the sewer systems were entirely carried by the wastewater flow and subsequently discharged into the river. This assumption was based on the local practice of street cleaning personnel, who typically gather larger waste items and deposit them into designated bins. Meanwhile, smaller items such as CBs are often swept into the sewer system along with dust and other particulate matter [60]. The concentration of different contaminants leached in the studied locations was estimated based on the multi-seasonal mean CBDs at those locations and the concentration leached from 1 CB to 1 L of water.

2.4. Health risk assessment

The potential contaminants leaching from CBs into the water include several inorganic (HMs) and organic chemicals [18,61–63]. The three exposure routes specified by the US Environmental Pollution Agency (USEPA) for various water contaminants are ingestion (oral consumption), dermal absorption (contact with skin), and inhalation (from inhalation of volatile compounds) [64]. Out of these, the ingestion route was selected for the current study as (i) the river water is the main drinking water source for the study area, (ii) dermal contact with contaminated water is less common and less significant compared to ingestion, and (iii) most of the CB leached contaminants are non-volatile, thereby eliminating the possibility of inhalation risk.

Health risk assessment was carried out in two steps: determination of CDI (mg/kg.day) followed by calculation of CR and NCR. The cancer and non-cancerous risks through ingestion were determined following the USEPA guidelines, which utilize the cancer slope factor (CSF) for CR assessment and reference dose (Rfd) for NCR assessment [64–66]. CDI was calculated using the following formula [67,68].

$$CDI_i = \frac{C_i \times IR \times EF \times ED \times CF}{BW \times AT} \quad (4)$$

where, CDI_i is the chronic daily intake of the ith contaminant (mg/kg.day), C_i is the concentration leached to water (μg/L), IR is the ingestion rate (L/day), EF is the exposure frequency (day/year), ED is the exposure duration (year), CF is the conversion factor from μg/L to mg/L, BW is the body weight (kg), and AT is the averaging time (day). CDI gives the daily intake of a contaminant, and it can be utilized to understand the lifetime intake (LTI) [67].

Once the CDI was determined, the CR was calculated using Eq. 5.

$$CR = \sum_{i=1}^{n} CDI_i \times CSF_i \quad (5)$$

where, CR is the total cancer risk (unitless, expressed as number of cases per million population), n is the number of contaminants considered, and CSF_i is the cancer slope factor as of the ith contaminant ((mg/kg.day)$^{-1}$).

Two indices, hazard quotient (HQ) and hazard index (HI) were utilized for the non-cancerous risk assessment. Initially established by the USEPA – National-Scale Air Toxics Assessment (NATA), HQ is defined as "*the ratio of potential exposure to a substance to the level at which no adverse effects are expected*," while HI is described as "*the cumulative sum of hazard quotients for toxins affecting the same target organ or organ system*" [69]. For instance, when evaluating a mixture of chemical compounds, HI represents the collective sum of HQs pertaining to an individual substance. The HQ and HI of the contaminants leached from CBs were evaluated using Eqs. 6 and 7.

$$HQ_i = \frac{CDI_i}{RfD_i} \quad (6)$$

$$HI = \sum_{i=1}^{n} HQ_i \quad (7)$$

where, HQ_i is the hazard quotient of the ith contaminant, RfD_i is the reference dose (mg/kg.day) of the ith contaminant, and HI is the sum of the hazard quotients. An HI value of 1 or lower suggests that adverse non-cancer health risks are improbable over a lifetime of exposure, rendering the hazards negligible. Conversely, an HI exceeding 1 indicates an elevated likelihood of adverse effects.

2.5. Model simulation and assumptions

Mathematical models are employed to predict exposure risks and serve as computational representations of real-life scenarios. The USEPA guidelines for human health exposure assessment advocate adopting probabilistic methodologies, such as Monte Carlo simulations, for model computations [64]. Accordingly, in the present study, 1,00,000 iterations of the Monte Carlo simulations of the models were carried out using Companion by MINITAB™ software. The model input parameters and their values were chosen based on past studies (Table 1). The overall variability in the model input parameters was accommodated through statistical distributions. These parameters include duration, population characteristics, and exposure-related variables, all exhibiting variation within and across individuals. Normal distribution was selected for the concentration of contaminants (C_i) as most studies reported it as such (Table 2). For other parameters, triangular distributions featuring the most probable values, along with minimum and maximum values, were selected. Many studies previously used triangular distribution for its ability to mitigate biases associated with potential outliers while effectively integrating variability into the model [70–72].

The model simulation for the WPPI and risk assessments were conducted based on the contaminant concentrations outlined in Table 2. These values denote the contaminants leached from a single CB into 1 liter of water after a 24-hour exposure period. Consequently, the initial simulation results provided the CDI, HI, and CR corresponding to a concentration of 1 CB/L, which can be universally applied.

In the subsequent modelling phase, data gathered from street CBD surveys and sewage discharge from Aizawl city were integrated to assess the actual CDI, HI, CR, and NCR considering the dilution effects. The sewers were considered to carry 80 % of the supplied water as per the Indian central pollution control board (CPCB) [76]. Consequently, the final model results would indicate the potential CDI, HI, CR, and NCR resulting from the total number of CBs in each sampling location. The sewage discharge is mostly drained into the Tlawng River, upstream of the intake point for the drinking water treatment plant [52]. Once released into the river, the sewage undergoes further dilution, thereby reducing the associated health risks.

Table 1
Model input parameters, distribution, and input values used in the current study [67,68,70–75].

Input parameter abbreviation	Name of input parameter and unit	Distribution and values
C_i	Concentration of contaminant	(Table 2)
IR	Ingestion rate (L/day)	T[a] (1, 2, 3)
EF	Exposure frequency (day/year)	T (330, 350, 365)
ED	Exposure duration (year)	T (68.3, 71.05, 73.8)
CF	Conversion factor	0.001
BW	Body weight (kg)	T (50, 60, 70)
AT	Averaging time (year)	T (24929.5, 25933.25, 26937)
CSF	Cancer slope factor ((mg/kg.day)$^{-1}$)	(Table 2)
RfD	Reference dose (mg/kg.day)	(Table 2)

[a] Triangular distribution.

Table 2
Contaminants leached from CBs to water and their concentrations.

Class	Contaminants	Concentration in mean ± standard deviation (μg/unit)[a]	Cancer slope factor (mg/kg-day)$^{-1}$ [b,c]	Oral reference dose (mg/kg-day)[b,c]	Weight of evidence (WOE)[b,c]	Maximum contamination limits in drinking water[d,e] (μg/L)	Reference
	Nicotine	2130 ± 106.5 2710 ± 135.5	-	-	-	-	[18] [104]
Heavy metal(loid)s	Arsenic	0.06 ± 0.003	1.5	3×10^{-4}	A[f]	10	[11]
	Cadmium	0.047 ± 0.002 0.115 ± 0.00575	-	5×10^{-4}	B1[g]	5	[62] [105]
	Nickel	0.255 ± 0.01275	-	2×10^{-2}	-	20	
	Copper	0.0058 ± 0.0003 0.55 ± 0.0275	-	4×10^{-2}	D[h]	2000	
	Zinc	0.021 ± 0.001 6.35 ± 0.3175	-	3×10^{-1}	D	-	
	Barium	0.0313 ± 0.0009	-	2×10^{-1}	D	2000	
	Aluminium	0.047 ± 0.002	-	1	-	200	
	Mercury	< LOD[i] (0.006 μg/L)	-	3×10^{-4}	D	1	
	Chromium	0.00259 ± 0.00002 0.1 ± 0.005	0.5	3×10^{-3}	_	25	
	Selenium	0.00354 ± 0.00001	-	5×10^{-3}	D	20	
	Manganese	17.3 ± 0.865 17.7 ± 0.885	-	1.4×10^{-1}	D	50	
	Lead	0.06275 ± 0.0031375 0.485 ± 0.02425	8.5×10^{-3}	-	B2[j]	5	
PAHs (Polycyclic Aromatic Hydrocarbons)	Naphthalene	0.02052 ± 0.01254 0.0045035 ± 0.0000535	0.12	2×10^{-2}	C[k]	0.1	[61] [105]
	Acenaphthylene	0.00456 ± 0.00114 0.0006495 ± 0.00002325	-	-	D	0.1	
	Acenaphthene	0.003002 ± 0.00076 0.001006 ± 0.000024	-	6×10^{-2}	-	0.1	
	Fluorene	0.03192 ± 0.01026 0.0007205 ± 0.0000165	-	4×10^{-2}	D	0.1	
	Anthracene	0.0285 ± 0.00038 0.0000215 ± 0.0000005	-	3×10^{-1}	D	0.1	
	Phenanthrene	0.00266 ± 0.00038 0.00077025 ± 0.0000303	-	-	D	0.1	
	Fluoranthene	0.00418 ± 0.00057 0.000235 ± 0.00003125	-	4×10^{-2}	D	0.1	
	Pyrene	0.00399 ± 0.00038 0.000221 ± 0.0000025	-	3×10^{-2}	D	0.1	
	Chrysene	< LOD (0.2 ng/L)	1×10^{-3}	-	B2	0.1	
	Benzo[*b*]fluoranthene	< LOD (0.2 ng/L)	0.1	-	B2	0.1	
	Benzo[*k*]fluoranthene	< LOD (0.4 ng/L)	0.01	-	B2	0.1	
	Benzo[*a*]pyrene	< LOD (0.5 ng/L)	1	3×10^{-4}	A	0.01	
	Indeno[*1,2,3-cd*]pyrene	< LOD (0.8 ng/L)	0.1	-	B2	0.1	
	Dibenz[*a,h*]anthracene	< LOD (0.8 ng/L)	1	-	B2	0.1	
	Benz[*a*]anthracene	0.000196 ± 0.0000035	0.1	-	B2	0.1	
BTEX (Benzene, Toluene, Ethylbenzene, Xylene) compounds	Benzene	0.0014 ± 0.00106	1.5×10^{-2}	4×10^{-3}	A	1	[106]
	Toluene	0.0056 ± 0.00392	-	8×10^{-2}	-	1	
	Ethylbenzene	0.0014 ± 0.00081	1.1×10^{-2}	1×10^{-1}	D	700	
	p-xylene	0.0015 ± 0.00079	-	2×10^{-1}	-	10,000	
	o-xylene	0.0011 ± 0.00049	-	2×10^{-1}	-	10,000	
Aromatic amines	Aniline	0.00875 ± 0.000493	5.7×10^{-3}	7×10^{-3}	B2	-	[103]
	1-Naphthylamine	0.00039 ± 0.000227	-	-	-	-	
	2-Naphthylamine	0.000119 ± 0.000031	1.8	-	-	-	
	2,6-dimethylaniline	0.0000812 ± 0.0000147	-	-	-	-	
	∑TOL (ortho-toluidine, Para-toluidine, meta-toluidine)	0.000064 ± 0.0000142	1.6×10^{-2} (o-toluidine) 3×10^{-2} (p-toluidine)	2×10^{-2} (o-toluidine) 4×10^{-3} (p-toluidine)	-	-	
	ortho-Anisidine	0.0000613 ± 0.0000423	1.4×10^{-1}	-	-	-	
	∑ABP (3-Aminobiphenyl, 4-Aminobiphenyl)	0.00000395 ± 0.00000034	21	-	-	-	

[a]Concentration after an exposure duration of 24 h; [b]USEPA IRIS (Integrated Risk Information System) Assessment; [c]RAIS (Risk Assessment Information System); [d]European Union Drinking Water Directive; [e]USEPA National Primary Drinking Water Regulations (NPDWR); [f]carcinogenic to humans; [g]probable human carcinogen

with limited human evidence; [h]not classifiable as to human carcinogenicity; [i]Limit of Detection; [j]probable human carcinogen based on sufficient evidence in animals and inadequate or no evidence in humans; [k]possible human carcinogen.

3. Results and discussion

3.1. CBs littering density (CBD)

The variation of CBD in the study locations across Aizawl City is depicted in Fig. 2. A total of (31,254 ± 3486) CBs were visually observed considering all the survey locations on an annual average (Table S2). This corresponds to an annual average CBD of (0.438 ± 0.669) CBs/m^2, ranging from 0.149 to 0.633 CBs/m^2. Analysis of CBD variation with land use indicated that commercial areas have the highest litter density, followed by institutional and residential areas. Location C6 (Bara Bazaar) has been identified as the primary hotspot with the highest annual average CBD. This is likely due to its central location, offering high accessibility and significant commercial activity, in addition to the presence of a daily vegetable market and several cafes in the vicinity. Further, locations I3, C4, and C5 also depicted high annual average CBDs of (0.628 ± 0.171), (0.624 ± 0.146), and (0.613 ± 0.335) CBs/m^2, respectively. The distribution of CBD across different commercially active public spaces, including malls and markets (29 %), shops (15 %), motor workshops (15.5 %), ATMs (12.6 %), bus stands (10.4 %), parking areas (7.85 %) and taxi stands (7.15 %) is depicted in Figure S3a.

The high CBDs of institutional areas, as observed in the case of I3 (Raj Bhavan) can be due to the presence of numerous government offices with high public flow in the vicinity. The distribution of CBD among institutional spaces followed the order of educational institutions (31.38 %), govt. offices (30.42 %), hospitals (28.25 %), and others (religious places, stadiums) (10.15 %) (Figure S3b). This is consistent with the findings of Valiente et al. [29], who reported that public places like hospitals, markets, and public transportation stops are the main littering hotspots for discarded smoked CBs. Previous studies have also demonstrated that high-density public spaces, particularly those related to retail, food services, and transportation hubs, consistently show higher levels of CB litter [31,77]. This can be attributed to several factors, including higher foot traffic, increased smoking frequency in leisure spaces (e.g., cafes), and insufficient waste management facilities [78,79]. Further, the behavioral study by Lehner et al. [80] on CB littering patterns identified high visitor turnover and transient behaviors, such as people smoking while moving between activities at urban markets, as key reasons for high CBD in commercial spaces. Other studies have also reported average urban area CBDs of 0.23 CBs/m^2 (Behbahan, Iran) [33], 0.25 ± 0.17 CBs/m^2 (Santos, Brazil), 0.08 ± 0.21 CBs/m^2 (Niteroi, Brazil) [31], 0.174 CBs/m^2 (Yasouj, Iran) [34], 0.21 CBs/m^2 (Qazvin, Iran) [42], 0.0056–1.16 CBs/m^2 (Hanoi, Vietnam) [36] and 0.005 – 0.385 CBs/m^2 (Jiroft, Iran) [32].

The seasonal variation of CBD across different locations is depicted in Table S3. The winter season depicted the highest CBD (0.551 ± 0.062 CBs/m^2), followed by summer (0.496 ± 0.066 CBs/m^2) and the rainy season (0.268 ± 0.045 CBs/m^2). The higher prevalence of CB littering

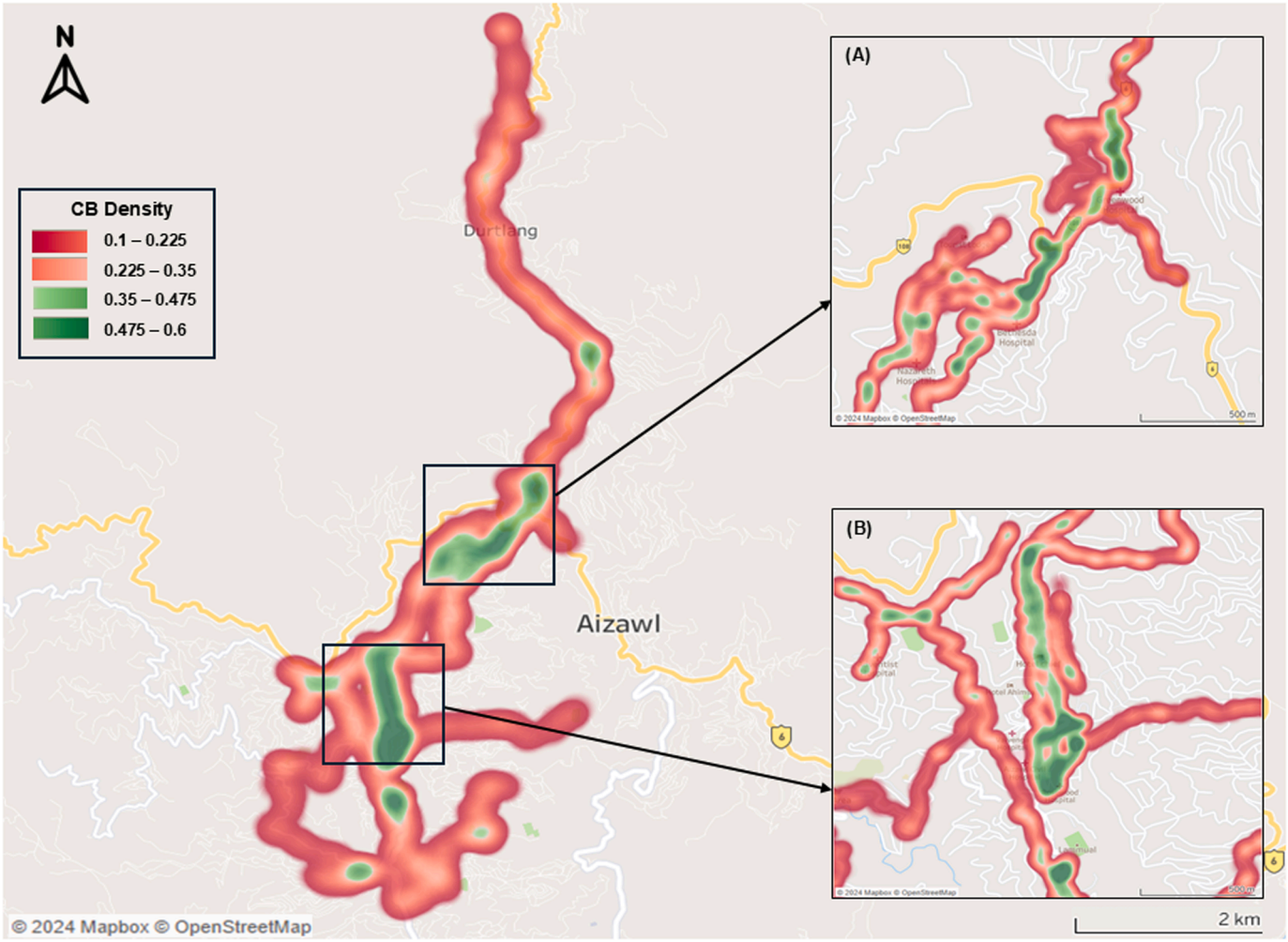


Fig. 2. CB littering density in Aizawl city (A) C2 (Bawngkawn), (B) C6 (Bara Bazar).

during the winter can be attributed to the moderately cold weather conditions in hilly areas like Aizawl, where smoking can provide a sense of comfort and relaxation [81,82]. A significant seasonal difference of 51.36 % was observed in the average CBD between the winter and rainy seasons. This variance may be due to the individuals' inclination to smoke indoors during the rainy season or the likelihood that a substantial portion of littered CBs is carried away by stormwater runoff along surface sewers.

Smokers' attitudes, behavior and beliefs towards littering play a major role in understanding the CB littering behavior of the smokers, which determines the spatial and temporal distribution of CBD in the studied areas [1]. Various factors including the presence/lack of CB receptacles, smoking locations, lack of accountability, convenience, lack of environmental awareness, quality of cleanup services, and public smoking regulations greatly influence CB littering [33,83]. Population density and standard of living can also be influential parameters in urban areas [37]. Moreover, socio-cultural and behavioral conformity to smoking norms will be another significant factor influencing CB littering in populations characterised by high smoking rates [84,85]. A detailed field-based interaction survey is necessitated to evaluate the knowledge, attitudes, and awareness of CB pollution in the city to determine the behavioral, socio-cultural, and infrastructural factors influencing their littering behavior. For instance, significant variation in CB littering behaviors can exist between distinct categories of smokers within a community, delineated by their smoking intensity (heavy vs light/social), which influences their choice of smoking locations, aiding in understanding the spatial distribution of CB littering in an area. Such insights can be instrumental in devising strategies to address social and individual barriers to smoking, consequently mitigating associated littering behaviors within societies where smoking is prevalent and less stigmatized.

3.2. Water pollution potential from CBs

Table 2 summarizes different organic and inorganic contaminants leached from CBs and their respective concentrations (mean ± SD) in μg/unit per 1 L of water. For uniformity, SD was assumed to be 5 % of the mean, if it was not provided in the study. Further, their respective CSF $(mg/kg\text{-}day)^{-1}$, oral Rfd (mg/kg-day), weight of evidence (WOE) for cancer incidence, and maximum contamination limit (MCL) (μg/L) in drinking water as per various regulatory agencies are depicted in Table 2. A total of 12 HMs including As, Cd, Ni, Cu, Zn, Ba, Al, Hg, Cr, Se, Mn, and Pb were reported from various studies with concentrations ranging from 0.00354 to 17.3 μg/unit in 1 L of water. HMs leached the highest among other contaminants, which could be a factor of their relatively higher absorbability and retention by the filter during combustion [86]. Further, the other leached contaminant concentrations followed the order: PAHs > BTEX > AAs (Table 2).

Among HMs, As (0.06 μg/unit) and Cd (0.047–0.115 μg/unit) are depicted as a known carcinogen to humans, Pb (0.06275–0.485 μg/unit) is considered as probable human carcinogen [87]. The other HMs are not classified as carcinogenic; however, they might contribute to non-carcinogenic toxicity. For instance, excessive exposure to Ni and Cr can cause skin and respiratory issues [88–90]; Cu, Hg, and Mn may lead to neurological symptoms and kidney damage [91,92]; Zn and Ba can cause gastrointestinal distress [93,94]; Al and Mn may result in neurotoxicity [95,96]; and Se can lead to hair loss and neurological issues [97]. Interestingly, the Ni and Pb concentrations in the scalp hair of steel production workers depicted by Afridi et al. [98] were comparable to our study. Further, the HM concentrations (Pb, Cd, Cu, and Cr) leached from 1 CB/L were similar to the HM levels in blood, hair, and nail samples of workers exposed in brick kilns [99].

The organic contaminants leaching from CBs include nicotine, PAHs, BTEX compounds, and AAs. The toxic, mutagenic, or carcinogenic effects of PAHs on human health are widely recognised [100]. Especially fluorene, anthracene, and naphthalene were reported to have high leachable concentrations. While benzo[*a*]pyrene is considered a known carcinogen, chrysene, benzo[*b*]fluoranthene, benzo[*k*]fluoranthene, indeno[*1,2,3-cd*]pyrene, dibenz[*a,h*]anthracene, naphthalene, and benz[*a*]anthracene are possible/probable human carcinogens. Within the group of BTEX compounds, known for their toxicity, stability, and cumulative effects [101], Benzene (0.014 μg/unit) is classified as a human carcinogen with the potential to elevate the incidence of leukemia and aplastic anemia [102]. Further, among the aromatic amines, aniline is classified as a potential carcinogen having the highest solubility in water with high concentration levels (0.00875 μg/unit) [103].

The WPPI calculation was carried out using the observed CBD in study locations. The proposed WPPI for CBs is based on the maximum contamination level in drinking water recommended by the EU Drinking Water Directive and the USEPA National Primary Drinking Water Regulations (NPDWR) [107,108]. In the present study, 24 parameters were used to evaluate the pollution level in surface water from CB. Among the parameters, Manganese exhibited the highest pollution potential (0.310 ± 0.048), while o-xylene depicted the lowest (1.296 ± 0.649) $\times 10^{-07}$. The Pollution Load for all the parameters were found in the order of Manganese > Naphthalene > Fluorene > Anthracene > Lead > Pyrene > Acenaphthylene > Acenaphthene > Fluoranthene > Cadmium > Phenanthrene > Nickel > Toluene > Arsenic > Benz[*a*]anthracene > Benzene > Chromium > Aluminium > Selenium > Copper > Barium > Ethylbenzene > p-xylene > o-xylene (Table S4). Therefore, integrating the individual pollution potentials of all contaminants leached from CBs, WPPI for a single CB in 1 L of water calculated using Monte Carlo Simulation was determined to be 5.2207×10^{-02}.

Further, WPPI per m^2 area of sampling locations was calculated by multiplying the WPPI for 1 CB/L of water to their respective mean CBD. The resulting average WPPI per m^2 of the study locations in Aizawl city was determined to be (0.022 ± 0.009). Consistent with the CBD distribution, commercial areas exhibited the highest WPPI (0.0297), followed by institutional (0.0197) and residential areas (0.0187). The distribution of WPPI per m^2 across the study areas is illustrated in Figure S4. Moreover, to determine the pollution potential of all the CBs littered, the WPPI for respective study locations to 1 L of water was calculated by multiplying the WPPI per CB per L of water by the total number of CBs littered in the area. This resulted in a total WPPI of (1631.69 ± 177.97), depicting the pollution potential of all the CBs littered across Aizawl to 1 L of water (Table 3). Similarly, the actual WPPI was calculated by considering the dilution of sewage produced by the city (equivalent to 80 % of 34 MLD). This yielded a value of (59.97 ± 6.68) $\times 10^{-06}$, which represented the total pollution potential of all the CBs littered in the city to the sewerage system. Hossain and Patra [57] have suggested that WPI < 0.5 signifies excellent water quality. However, as the present study solely focuses on potential water pollution from littered CBs, comparisons with other sources are not feasible. Instead, it can be inferred that littered CBs alone contribute to (59.97 ± 6.68) $\times 10^{-06}$ of the total WPI of sewage, which further contaminates the surface water downstream.

Considering further dilution of sewage after mixing with the surface water and stormwater, the water pollution potential of CBs to the city's surface water source is expected to decrease more significantly. It should be noted that in the context of the current study, the lack of reliable data regarding stormwater generation and the surface water runoff of the city limits the accurate calculation of CBs' contribution to water pollution after dilution, which can be considered a limitation of the study. Nevertheless, the findings of WPPI suggest that CBs significantly contribute to water pollution alongside contaminants from other sources. Moreover, the WPPI calculated for a single CB/L of water can be used to accurately calculate the water pollution potential of CB littering in a city for any specific water source.

3.3. Human health impacts of CBs

3.3.1. Potential chronic daily intake of contaminants from CB littering

The CDI calculation considered all contaminants with concentrations

Table 3
Water pollution potential (mean ± SD) from CBs littered across Aizawl city estimated through water pollution potential index (WPPI).

Type of area	Sampling area	WPPI per m^2 area to 1 L water	WPPI for study locations to 1 L water	Actual WPPI after dilution with sewage ($\times 10^{-06}$)
Institutional	I1	0.021 ± 0.007	81.13 ± 27.26	2.98 ± 1.00
	I2	0.008 ± 0.004	26.83 ± 13.20	0.99 ± 0.48
	I3	0.033 ± 0.009	84.94 ± 23.15	3.12 ± 0.85
	I4	0.017 ± 0.009	48.71 ± 27.04	1.79 ± 0.99
Residential	R1	0.029 ± 0.011	193.17 ± 73.02	7.10 ± 2.68
	R2	0.022 ± 0.006	122.48 ± 32.46	4.50 ± 1.19
	R3	0.024 ± 0.01	156.52 ± 65.69	5.75 ± 2.41
	R4	0.014 ± 0.009	68.08 ± 25.67	2.51 ± 1.68
	R5	0.011 ± 0.005	75.23 ± 37.93	2.76 ± 1.39
	R6	0.012 ± 0.005	64.48 ± 25.50	2.37 ± 0.94
Commercial	C1	0.030 ± 0.008	120.13 ± 32.93	4.42 ±1.21
	C2	0.025 ± 0.007	105.93 ± 31.49	3.89 ± 1.16
	C3	0.025 ± 0.004	138.56 ± 23.50	5.09 ± 0.86
	C4	0.033 ± 0.008	64.68 ± 15.169	2.38 ± 0.56
	C5	0.032 ± 0.017	205.75 ± 112.58	7.56 ± 4.14
	C6	0.033 ± 0.011	75.07 ± 22.57	2.76 ± 0.83
Total WPPI			1631.69 ± 177.97	(59.97 ± 6.68) × 10^{-06}

[a]Selected based on the total pollution index of the area.

Table 4
Chronic daily intake, hazard index, and cancer risk of potential contaminants from one smoked CB to 1 L of water.

Contaminant	CDI (mg/kg.day) ($\times 10^{-06}$)	Hazard quotient ($\times 10^{-06}$)	Cancer risk ($\times 10^{-06}$)
Heavy metals (HMs)			
Arsenic	1.969 ± 0.456	6563 ± 1518	2.952 ± 0.681
Cadmium	3.034 ± 1.022	6068 ± 2044	_
Nickel	8.284 ± 1.931	414.2 ± 96.53	_
Copper	7.137 ± 5.289	178.4 ± 132.2	_
Zinc	146.9 ± 62.59	489.8 ± 208.6	_
Barium	0.975 ± 0.232	4.877 ± 1.162	_
Aluminium	1.257 ± 0.348	1.527 ± 0.348	_
Chromium	1.327 ± 0.947	442.4 ± 315.5	0.665 ± 0.476
Selenium	0.113 ± 0.025	22.68 ± 4.955	_
Manganese	494.8 ± 133.9	3534 ± 956.5	_
Lead	11.73 ± 4.513	_	0.099 ± 0.038
Polycyclic aromatic hydrocarbons (PAHs)			
Naphthalene	0.801 ± 0.557	40.05 ± 27.86	0.096 ± 0.066
Acenaphthene	0.087 ± 0.054	1.444 ± 0.902	_
Fluorene	0.750 ± 0.560	18.74 ± 14.01	_
Anthracene	0.618 ± 0.260	2.059 ± 0.866	_
Fluoranthene	0.070 ± 0.049	1.738 ± 1.221	_
Pyrene	0.109 ± 0.046	3.647 ± 1.534	_
Benz[*a*]anthracene	0.006 ± 0.001	_	0.0006 ± 0.000
BTEX compounds			
Benzene	0.060 ± 0.045	15.01 ± 11.32	0.001 ± 0.000
Toluene	0.336 ± 0.203	4.201 ± 2.539	_
Ethylbenzene	0.082 ± 0.043	0.819 ± 0.428	0.001 ± 0.000
p-xylene	0.093 ± 0.043	0.463 ± 0.216	_
o-xylene	0.041 ± 0.023	0.206 ± 0.115	_
Aromatic amines (AAs)			
Aniline	0.278 ± 0.066	48.79 ± 11.50	0.001 ± 0.000
2-Naphthylamine	0.015 ± 0.010	_	0.027 ± 0.017
∑TOL[a]	0.002 ± 0.001	0.132 ± 0.065	3.4 × 10^{-05} ± 0.0
ortho-Anisidine	0.003 ± 0.002	_	0.0004 ± 0.000
∑ABP[b]	0.0001 ± 0.000	_	0.002 ± 0.001

[a] ortho-toluidine, Para-toluidine, meta-toluidine; [b]3-Aminobiphenyl, 4-Aminobiphenyl.

above LOQ (Table 2). The results of CDI analysis for 1 CB/L are summarized in Table 4. Similar to the leached concentration of different classes of contaminants, the order of each contaminant category to the total CDI was HMs > PAHs > BTEX ≈ AA, with over 99 % attributed to HMs. Corresponding to the leached concentration, Mn ((494.8 ± 133.9) $\times 10^{-06}$ mg/kg.day from 1 CB/L) posed the highest CDI (73.03 %) in HM-CDI, followed by Zn (21.68 %), Pb (1.73 %), and Ni (1.22 %). For comparison, in a study on contaminated lake water ingestion, the highest CDI was also from Mn (0.071 mg/kg.day), followed by Fe, Zn, Cu, Ni, and As - substantially higher than the values observed in the present study [109]. Similarly, the CDI (through ingestion and dermal pathways) of HMs in the groundwater of Qatar showed that Mn had the highest contribution, resulting in a total HM-CDI of 1.4 × 10^{-05} – 6.7 × 10^{-01} mg/kg.day [110]. The potential CDI of total CBs littered in the Aizawl city is illustrated in Fig. 3a. While the commercial and residential areas contributed similarly to CDI, the institutional areas exhibited less potential.

The CDI measurement helps understand the lifetime intake of contaminants. The CDI of 1 CB/L can result in a lifetime intake of 1.126 g of HMs, 4.05 mg of PAHs, 1.019 mg of BTEX, and 0.465 mg of AA for a life expectancy of 70 years and a body weight of 65 kg. This is based on drinking water contaminated at the rate of 1 CB/L, which might reduce as the dilution increases. For instance, in Aizawl city, the lifetime intake of CB-leached contaminants can vary between 21.41 and 164.1 μg (mean = 81.36 μg) for the same lifetime and body weight. The lifetime intake increases as the lifetime exposure increases. The average daily intake of Pb, Cd, Ni, and Zn by smokers (26–40 years of age) through inhaling mainstream smoke was found to be 3.06 $\times 10^{-09}$, 2.48 $\times 10^{-10}$, 1.37 $\times 10^{-09}$, and 7.88 $\times 10^{-09}$ mg/kg. day, respectively [111]. These values correspond to a lifetime intake of 5.082 (Pb), 0.412 (Cd), 2.275 (Ni), and 13.08 (Zn) μg, making a total of 20.85 μg. This result suggests that at the lowest CBD in Aizawl city, the lifetime intake is similar (21.41 μg as opposed to 20.85 μg) to the lifetime intake of HMs through smoking.

The residual tobacco within the cigarette, the contaminants trapped in the filter from mainstream smoke, and the filter material itself are the primary sources of contaminants. The risks to human health posed by HMs involve exposure to Pb, Cd, and As, which can accumulate in the body, leading to health issues in children and adults. Additionally, trace amounts of essential HMs like Cr and Ni are vital but can become harmful when present in excessive concentrations [112]. Further, the Cd and Pb concentrations in blood samples of smokers showed a significant positive association with chronic obstructive pulmonary disease [113]. Therefore, the CDI and lifetime intake can be considered indicators for exposure and subsequent health risks. The 5th, 50th and 95th percentile of CDIs of all the contaminants through the ingestion route were found to be 4.14 × 10^{-04}, 6.71 × 10^{-04}, and 9.92 × 10^{-04}, respectively.

3.3.2. Non-cancerous risk associated with CB littering

NCR assessment is conducted for those contaminants with oral reference doses and concentrations above LOQ. The total HI calculated for all the contaminants was 1.785 × 10^{-02} based on a single CB, which is less than 1, indicating no potential NCR. While an individual CB did not exhibit any noticeable NCR effect, accumulated effects of a significant number of CBs could potentially result in NCR concerns. As the precise quantity of littered CBs that ultimately reach downstream water bodies remains unknown, the study was intended to compare levels of exposure rather than conducting an actual risk assessment. This necessitates giving priority to non-cancer adverse health effects from CBs [114]. The CDI, HQ for various contaminants, and HI for different

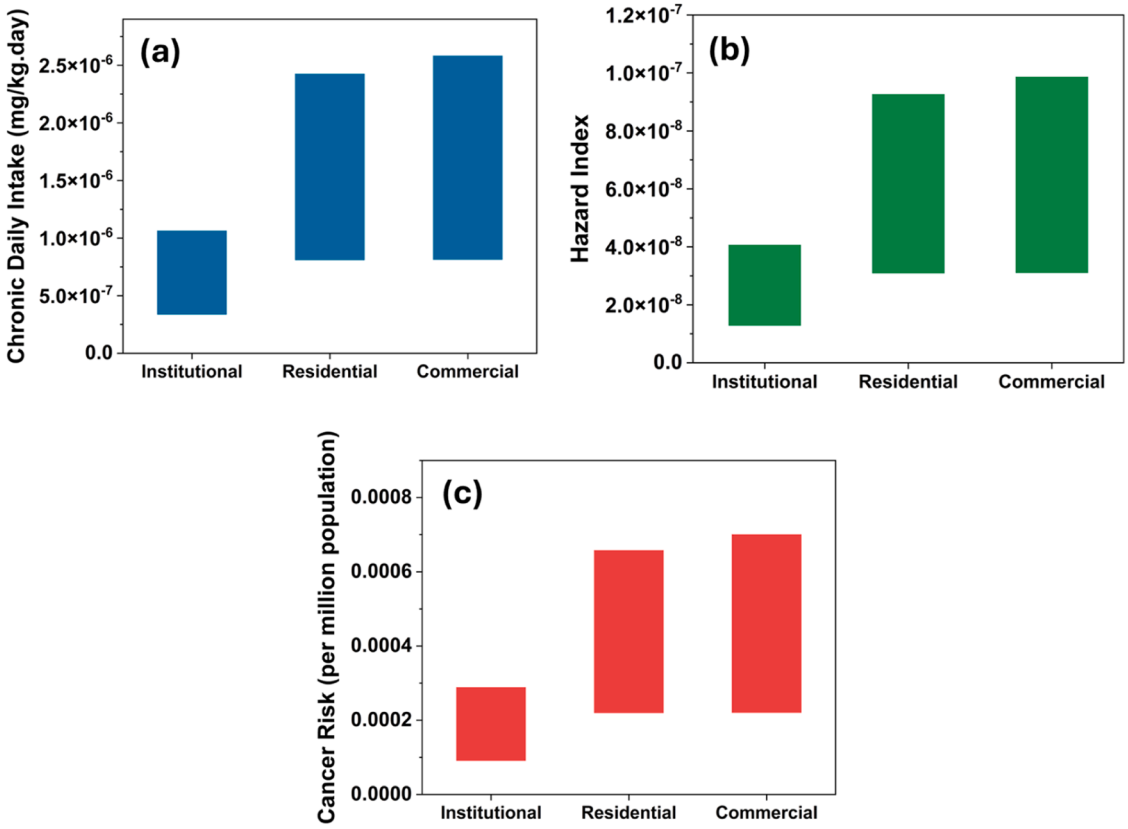


Fig. 3. (a) Chronic daily intake, (b) hazard index, and (c) cancer risk of contaminants leached from littered CB at various locations in Aizawl city.

location types are shown in Table 4 and Fig. 3.

Among the various contaminants, As posed the highest HQ of 6.563 $\times 10^{-03}$ (36.75 % of total HQ), followed by Cd (33.98 %), Mn (19.79 %), Zn (2.74 %), Cr (2.47 %), and Ni (2.31 %). The findings indicated that HMs have the highest contribution, with 99.2 % of the total CDI from all contaminant groups. The different trend of HQ from CDI is due to the relatively lower RfD values of contaminants. Ahmed et al. [111] found Cd, Zn, and Ni HQ to be 1.81 $\times 10^{-04}$, 7.68 $\times 10^{-05}$, and 1.87 $\times 10^{-04}$, respectively, for smoking workers in the plastic industry. Our findings demonstrated the HQ value for the same as 60.68 $\times 10^{-04}$, 4.89 $\times 10^{-05}$, and 4.14 $\times 10^{-04}$ respectively, indicating a substantial increase compared to the previous study [111].

A key observation of the current study is regarding the leached nicotine concentration. Nicotine has well-documented health effects, including addictive properties and cardiovascular impacts [115]. Smoking one cigarette is estimated to cause an intake of ~1.25 mg of nicotine per cigarette [116,117]. Studies reported up to a concentration of 2710 μg/CB/L of nicotine leaching (Table 2), which can result in a CDI of (7.55 ± 1.95) $\times 10^{-02}$ mg/kg.day. Although this intake is lower than from smoking, it is concerning because exposure to drinking water is unavoidable, unlike the voluntary habit of smoking.

3.3.3. *Cancer risk associated with CB littering*

Contaminants with established CSF values by USEPA and concentrations above LOQ were included in the CR assessment. The findings are summarized in Table 4 and depicted in Fig. 3c. The result summarized in Table 4 is for one smoked CB/L of water and it can be used to determine the total CR at different locations if the CBD data is available. The total CR calculated was 3.847 $\times 10^{-06}$ from 1 CB/L considering all contaminants. This value corresponds to a potential CR of 3.84 people out of a million population. Out of all contaminants, As induced the highest risk (76.77 %). The order of the next five highest contributors to total CR was Cr (17.29 %) > Pb (2.57 %) > Naphthalene (2.49 %) > 2-Naphthylamine (0.70 %) > ∑ABP (0.05 %).

The total CR from all four contaminant groups after considering the dilution was negligible (< 1 $\times 10^{-06}$ or one in a million population) (Fig. 3c). Like CDI, HMs posed the highest CR potential (96.5 %), followed by PAHs (2.51 %), AAs, and BTEX compounds. The decrease in the contribution of HMs to CDI and CR (99 % of total CDI constituted HMs, while only 96 % of CR originated from HMs) could be attributed to i) the CSF values associated with each contaminant and ii) the number of contaminants contributing to the total CDI and CR (Table 2 and Fig. 3).

The CR observed in the current study is comparable to findings from other studies, though generally lower than many reported values. For instance, Badeenezhad et al. [118] observed a higher CR (9.96 $\times 10^{-05}$) value just for Cr, suggesting more significant risks in residential drinking water of Iran. Similarly, CR values of HMs including Cr^{6+} and As (up to 1 $\times 10^{-04}$), PAHs (anthracene and fluoranthene) (7.38 $\times 10^{-10}$) which were notably higher than those observed in this study [119–123]. Therefore, while the contaminants in CBs contribute to a notable CR, particularly As, the values are lower than those reported in studies focusing on HMs and PAHs in contaminated water sources.

The CR assessment results of the current study can be further compared with the CR potential linked to cigarette smoking. A multinational data analysis on 158 chemical compounds originating from cigarette smoking was conducted by Fowles and Dybing [114]. They found that smoking one cigarette daily over a lifetime (60 years) could result in a CR of 1.16 $\times 10^{-04}$ (116 per million) solely from As. Comparatively, the CR associated with As leached from a single smoked CB into 1 L of water after 24 hours was found to be 2.952 $\times 10^{-06}$ (2.95 per million) (Table 4), which is 1.98 % of the risk of lifetime smoking. Similarly, the CR of other prevalent contaminants leached from CBs, namely, Cr, Pb, 2-Naphthylamine, and ∑ABP were approximately 2.11 %, 1289 %, 15 %, and 0.56 %, respectively, of the risk associated with lifetime smoking. Another study reported that the HM concentration (Pb, Cd, and Cr) in different cigarettes sold in Bangladesh can result

in a maximum CR of 1.48×10^{-02} to 2.27×10^{-01} (14,800–227,000 per million) [124]. The total CR associated with the contaminants in mainstream smoke was found to be 9.30×10^{-04} (930 per million) [114]. The total CR from one smoked CB to 1 L was observed as 3.847×10^{-06} in the present study, which is 0.414 % of the lifetime smoking of one cigarette per day. Therefore, the lifetime CR of 242 CBs/L in water would correspond to smoking 1 cigarette per day. The 5th, 50th and 95th percentiles of total lifetime risk of cancer associated with exposure to CBs-related contaminants for the oral ingestion route were 2.357×10^{-6}, 3.773×10^{-6}, and 5.599×10^{-6} respectively. Table S5 summarizes the 5th, 50th and 95th percentile values of total lifetime CR and CDI for various contaminants related to CBs.

3.4. Limitations and future scope

It is important to note that the dilution after discharging into the river has not been factored into this study, which is acknowledged as a limitation. Data limitations on estimated quantities of sewage and surface runoff from Aizawl city restricted the analysis of the overall potential for surface water pollution in the river. As a result, the findings can be primarily interpreted as the potential health risks CBs pose to the population in city sewage, as reflected by the WPPI used in the current study. However, with comprehensive data, the models and assessment could be extended to surface water sources receiving sewage, stormwater, and surface runoff. Further, the study did not include an ecological risk assessment due to the lack of standardized toxic response factors for many emerging contaminants such as PAHs, BTEX and several others leached from the CBs. Future research could address this gap by developing and standardizing toxic response factors.

4. Conclusion

The study revealed that the average annual CBD in Aizawl city was (0.438 ± 0.669) CBs/m^2, with several commercial (markets, shops, motor workshops, ATMs) and institutional (hospitals, schools, and government buildings) locations identified as primary hotspots. The water pollution potential index (WPPI) for a single CB/L of water was determined to be 5.22×10^{-02}. Extrapolating to the city's total CB litter, the WPPI indicates a cumulative impact on the municipal sewerage system of $(59.97 \pm 6.68) \times 10^{-06}$, underscoring a modest yet notable contribution to surface water pollution. Utilizing the established WPPI for 1 CB/L facilitates precise quantification of water pollution potential from CB littering to any specific water source, given robust data on sewage and surface runoff. Furthermore, the WPPI framework can extend to assess diverse pollutant sources, enhancing understanding of their relative impacts on water quality, which can be a promising scope of future research. The health risk assessment depicted HMs leached from CBs as a principal contaminant class of concern with a lifetime exposure potential of 1.126 g from 1 CB/L. This further contributed to the cumulative CR of 3.847×10^{-06} and HI of 1.785×10^{-02}, depicting significant cancerous and non-cancerous risks to the population. Overall, it was calculated that the lifetime CR associated with 242 CBs/L in water was equal to smoking 1 cigarette per day. The WPPI and human health risk models developed in this study can be adapted to any location (point/area) with relevant localized input data on littered CBs, concentrations of leached contaminants, water dilution, and population statistics. Considering the complexity of CB littering as a multifaceted issue, it is essential to comprehensively analyze all the behavioral, sociocultural, and infrastructural factors influencing CB littering behavior to design effective, targeted solutions. Further, adopting an integrated systems approach, encompassing infrastructural enhancements, regulatory policies, and community engagement initiatives, will be crucial for effectively mitigating CB pollution and its associated risks.

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CRediT authorship contribution statement

Kumar Raja Vanapalli: Writing – review & editing, Visualization, Supervision, Resources, Project administration, Funding acquisition, Conceptualization. **Naseeba Parveen:** Writing – original draft, Visualization, Conceptualization. **Lourembam Nongdren:** Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.jece.2024.114941.

Data availability

Data will be made available on request.

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